Machine Learning Case Study

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Problem Statement: Customer Purchase Predictions using Logistics Regression and Random Forest

Background:

In the business and retail sector, predicting customer purchase behaviour is essential for improving sales and marketing strategies. Businesses use customer data to identify buying patterns and predict purchase decisions. This project applies Logistic Regression and Random Forest machine learning algorithms to a dataset containing 1,500 customer records to predict whether a customer will make a purchase. The performance of both models is evaluated and compared to identify their effectiveness in predicting customer purchase behaviour.

Importance of the Problem:

Customer purchase prediction helps businesses understand customer behaviour and improve marketing effectiveness using machine learning algorithms such as Logistic Regression and Random Forest.

- Identifies potential buyers based on customer data.
- Supports targeted advertising and personalized marketing strategies.
- Reduces unnecessary marketing expenses by focusing on likely buyers.
- Improves customer engagement, sales planning, and decision-making.
- Compares Logistic Regression and Random Forest to evaluate their effectiveness in predicting customer purchase behaviour.

Objective:

The main objective is to develop and compare Logistic Regression and Random Forest models to predict whether a customer will purchase a product based on customer-related features.

The project also aims to evaluate and compare both models using accuracy, precision, recall, F1-score, and a confusion matrix to determine their effectiveness in predicting customer purchase behaviour.

Challenges:

- Class Imbalance: Differences in purchase and non-purchase records may cause either model to favour the majority class.
- Complex Customer Behaviour: Purchase decisions depend on multiple factors, such as income, age, discounts, and shopping habits. Logistic Regression may struggle with complex non-linear relationships, while Random Forest can capture these patterns more effectively.
- Prediction Errors: False positives and false negatives can lead to ineffective marketing strategies and missed potential customers.
- Feature Dependence: The performance of both models depends on the quality, relevance, and proper preprocessing of the available customer data.
- Model Selection: Comparing Logistic Regression and Random Forest helps identify which model provides better predictive performance for customer purchase prediction.
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Applications:

Customer purchase prediction can be applied in:

- E-commerce: Identifying potential buyers.
- Digital marketing: Creating targeted advertising campaigns.
- Retail: Understanding customer buying patterns.
- Customer analytics: Supporting customer segmentation and sales planning.

Methodology:

Workflow and Strategy:

1. Logistic Regression Workflow

1. Upload the Kaggle dataset into Google Colab.
2. Explore the dataset and check for missing values and inconsistencies.

3. Separate the input features from the target variable.
4. Split the dataset into 80% training data and 20% testing data.
5. Standardize the numerical features using `StandardScaler` and encode categorical variables where required.
6. Train the Logistic Regression model using the training data.
7. Predict whether customers will purchase a product using the testing data.
8. Evaluate and visualize the results using accuracy, precision, recall, F1-score, and a confusion matrix.

2. Random Forest Workflow

1. Upload the Kaggle dataset into Google Colab.
2. Explore the dataset and check for missing values and inconsistencies.
3. Separate the input features from the target variable.
4. Split the dataset into 80% training data and 20% testing data.
5. Encode categorical variables where required. Feature standardization is generally not necessary for Random Forest.
6. Train the Random Forest model using multiple decision trees.
7. Predict whether customers will purchase a product using the testing data.
8. Evaluate and visualize the results using accuracy, precision, recall, F1-score, a confusion matrix, and feature importance.

Implementation:

- **Dataset Name:** Customer Purchase Dataset (https://www.kaggle.com/datasets/rabieelkharoua/predict-customer-purchase-behavior-dataset/data)
- **Target Variable:** `PurchaseStatus` (0 = No Purchase, 1 = Purchase)
- **Key Features:** Customer demographic details, income, shopping behaviour, and other customer-related attributes used to predict purchase decisions.
- **Model Application:** Logistic Regression estimates the probability of a customer making a purchase, while Random Forest combines multiple decision trees to identify patterns in customer data and predict purchase behaviour.

Python link:

Logistic Regression:
https://colab.research.google.com/drive/1YU1MFOmhG0mLxwpkaZKxA0h5wOB2m0F9?usp=sharing

Random Forest:

https://colab.research.google.com/drive/1axxa87_PZm8lRP15pF_Z_zl61DhdeUQW?usp=sharing

Github Link:

https://github.com/purvipatankar/customer-purchase-prediction-ml

Results Analysis and Display

The Logistic Regression model achieved 82.67% accuracy, while the Random Forest model achieved 93.33% accuracy on the test dataset of 300 records. Both models were evaluated using precision, recall, F1-score, and a confusion matrix to assess their effectiveness in predicting customer purchase behaviour.

1. Logistic Regression — Evaluation Results

Evaluation metric	No Purchase	Purchase
Precision	0.83	0.82
Recall	0.88	0.76
F1-score	0.85	0.79

The Logistic Regression model correctly classified 248 records and incorrectly classified 52 records. It performed better at identifying customers who would not make a purchase than those who would make a purchase.

2. Random Forest — Evaluation Results

The Random Forest model achieved 93.33% accuracy, correctly classifying 280 out of 300 records and incorrectly classifying 20 records.

Evaluation metric	No Purchase	Purchase
Precision	0.93	0.94
Recall	0.95	0.91
F1-score	0.94	0.92

The Random Forest model demonstrated strong performance for both classes. It identified 95% of actual non-purchasing customers and 91% of actual purchasing customers. Its ROC-AUC score was 0.942, indicating a strong ability to distinguish between customers who purchase and those who do not.

Evaluation Metric	Logistic Regression	Random Forest
Accuracy	82.67%	93.33%
Correct predictions	248/300	280/300
Incorrect Prediction	52/300	20/300
No Purchase — F1-score	0.85	0.94
Purchase — F1-score	0.79	0.92
ROC-AUC	-	0.942

The results show that Random Forest outperformed Logistic Regression on this test dataset, achieving an accuracy improvement of 10.66 percentage points and making 32 fewer classification errors. It also achieved higher precision, recall, and F1-scores for both purchase and non-purchase classes.

Strengths and Weaknesses Analysis:

Logistic Regression

- Strengths: Simple to implement, computationally efficient, and provides purchase probabilities for clear decision-making. It also serves as an interpretable baseline for comparing model performance.
- Weaknesses: May struggle to capture complex, non-linear relationships in customer behaviour and can make incorrect predictions when patterns are more complex.

Random Forest

- Strengths: Combines multiple decision trees to improve prediction accuracy, captures complex customer behaviour patterns, and provides feature importance scores to identify influential factors in purchase decisions.
- Weaknesses: Requires more computational resources than Logistic Regression and is less interpretable because it combines predictions from multiple decision trees.

Key Insights and Conclusions:

The project demonstrates the use of Logistic Regression and Random Forest for customer purchase prediction. Logistic Regression achieved 82.67% accuracy, while Random Forest achieved 93.33% accuracy, making Random Forest the better-performing model.

The project highlights the importance of data preprocessing, model evaluation, and visualization in predicting customer behaviour. Further improvements could include hyperparameter tuning and testing additional features to enhance prediction performance.

References:

1. **Dataset:** Kaggle — Customer Purchase Prediction Dataset.
2. **Scikit-learn:** Logistic Regression
3. **Scikit-learn:** Random Forest Classifier
4. **Scikit-learn:** Data Preprocessing and StandardScaler
5. **Scikit-learn:** Model Evaluation Metrics
6. **Google Colab:** Development Environment